

Project 9: Financial Data Collection and Cleaning Pipeline

Objective: Build a comprehensive financial data pipeline that fetches data from multiple sources, cleans it, and prepares it for analysis across different asset classes.

Requirements:

1. Create a modular data collection function that can fetch data from at least two different sources:
 - Yahoo Finance (using yfinance)
 - Another source like Alpha Vantage, FRED, or Quandl (your choice)
2. Build the pipeline to handle different types of financial data:
 - Stocks (at least 5 different companies from various sectors)
 - A market index (like S&P 500)
 - A cryptocurrency
 - A forex pair
3. Implement robust data cleaning procedures:
 - Detect and handle missing values using appropriate filling methods
 - Identify outliers using z-scores or other statistical methods
 - Adjust for stock splits and dividends
 - Standardize data across different assets for comparison
4. Add derived features that could be useful for trading:
 - Daily returns